

Spacecraft Subsystem Design Process

Chapter 11

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SECTION OUTLINE

1. Attitude Determination and Control Subsystem (ADCS) Design Process
 2. Telemetry, Tracking, and Command (TT&C)
 3. Command and Data Handling Subsystem (C&DHS)
 4. Power
 5. Thermal Control
 6. Structures and Mechanisms
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This chapter provides more detailed technical information about spacecraft subsystems that supports the overall spacecraft architecture and design discussed in [Chapter 10](#). Top-level requirements are discussed for each subsystem along with a brief discussion of the concept of operations, architecture and design trade-offs and important interfaces. Important hardware and software considerations are described along with examples of components and overall configuration.

First-order approximations of subsystem size, mass, and power are provided where possible, sufficient to estimate development cost. The propulsion subsystem discussion, along with a broad description of propulsion technologies, is provided in [Chapter 17](#). The communications subsystem, along with communications payload design and broad communications theory is described in [Chapter 13](#).

Attitude Determination and Control Subsystem (ADCS)

The *attitude determination and control subsystem (ADCS)* stabilizes the vehicle and orients it in desired directions during the mission despite the internal and external disturbance torques acting on it. This requires either that the vehicle be inherently stable about the desired attitude, or that the vehicle determine its attitude, using sensors, and control it, using actuators. The ADCS is often tightly coupled to other subsystems on board, especially propulsion and navigation.

We begin by discussing several useful concepts and definitions, including mass properties, disturbance torques, angular momentum, and reference vectors. The mass properties of a spacecraft are key in determining the size of control and disturbance torques. We generally need to know the location of the center of mass (c_m) as well as the elements of the inertia matrix: the moments and products of inertia about chosen reference axes. The directions of the principal axes—those axes for which the inertia matrix is diagonal and the products of inertia are zero—are also of interest. Finally, we need to know how these properties change with time, as fuel or other consumables are used, or as appendages are moved or deployed.

A body in space is subject to small but persistent disturbance torques (about 10^{-4} N·m or less) from a variety of sources. These torques are categorized as cyclic (varying in a sinusoidal manner during an orbit), or secular (accumulating with time, and not averaging out over an orbit). These torques would quickly reorient the vehicle unless resisted in some way. An ADCS system resists these torques either passively, by exploiting inherent inertia or magnetic properties to make the disturbances stabilizing and their effects tolerable, or actively, by sensing the resulting motion and applying corrective torques.

Attitude Determination and Control Subsystem (ADCS) Design Process

The primary objective of the attitude determination and control subsystem (ADCS) is to ensure that the spacecraft stabilizes about the desired attitude (or attitude rates), despite the presence of disturbance torques, electronic noise in the control system, and vibrations of the vehicle's structure. The overall process used to meet this objective, along with required inputs and outputs, is shown in [Table 11.1](#) .

Table 11.1 Attitude Determination and Control Subsystem (ADCS) Design Process. This iterative process guides the design of the ADCS. (Adapted from SMAD Table 11-1.)

Step	Inputs	Outputs	FireSAT Example
1. a. Define control modes b. Define or derive system-level requirements by control mode	Mission requirements, mission profile, type of insertion for launch vehicle	- List of different control modes during mission (See Table 11.3) - Requirements and constraints (See Table 11.4)	- Orbit injection mode: none—provided by launch vehicle - Normal operating mode: nadir pointing, $< 0.1^\circ$; autonomous determination (Earth-relative) - Optional slew mode: one 30° maneuver per month to a target of opportunity
2. Select type of spacecraft control by attitude control mode	- Payload, thermal, and power needs - Orbit, pointing direction - Disturbance environment	Method for stabilizing and control: 3-axis, spinning, or gravity gradient	- Reaction wheels for attitude control and slewing - Magnetic torquing for momentum dumping
3. Quantify disturbance environment	Spacecraft geometry, orbit, solar and magnetic models, mission profile	Values for forces from gravity gradient, magnetic field, aerodynamics, solar pressure, internal disturbances, and powered flight effects on control (c_m offsets, slosh)	- Gravity gradient: 1.8×10^{-6} N·m normal pointing; 4.4×10^{-5} N·m during target-of-opportunity mode - Magnetic: 4.5×10^{-5} N·m - Solar: 6.6×10^{-6} N·m - Aerodynamic: 3.4×10^{-6} N·m
4. Select and size ADCS hardware	Spacecraft geometry, pointing accuracy, orbit conditions, mission requirements, lifetime, orbit, pointing direction, slew rates	- Sensor suite: Earth, Sun, inertial, or other sensing devices - Control actuators, e.g., reaction wheels, thrusters, or magnetic torquers - Data processing electronics, if any, or processing requirements for other subsystems or ground computer	4 reaction wheels, 0.4 N·m·s capacity 2 horizon sensors, scanning, 0.1° accuracy 3 electromagnets, dipole moment: 30 A·m² 4 Sun sensors, 0.1° accuracy 1 three-axis magnetometer, 1° accuracy
5. Define determination and control algorithms	All of above	Algorithms, parameters, and logic for each determination and control mode	- Determination: horizon data filtered for pitch and roll. Magnetometer and Sun sensors used for yaw. - Control: proportional-plus-derivative
6. Iterate and document	All of above	- Refined requirements and design - Subsystem specification	

More detailed objectives for an Earth-orbiting spacecraft would likely include the following:

- ☀ Maintain spacecraft attitude pointing accuracy over the spacecraft’s entire lifetime
- ☀ Point payload instruments as required
- ☀ Point solar arrays for all stages of the operational lifetime
- ☀ Perform vehicle slewing as required
- ☀ Maintain antenna pointing for all necessary uplinks, downlinks, and crosslinks
- ☀ Keep vibration environment within requirements for all onboard systems
- ☀ Facilitate orbit maintenance and orbit-change maneuvers
- ☀ Integrate harmoniously with all other spacecraft subsystems
- ☀ Meet ADCS mass, power, and volume constraints

Table 11.2  provides examples of the sources of ADCS requirements and necessary information.

TABLE 11.2  **Sources of Attitude Determination and Control Subsystem (ADCS) Requirements.** Potential sources of ADCS requirements are shown here along with a brief definition of the information required. (Adapted from SMAD Table 10-13.)

Requirement	Information Needed
Payload requirements: • Article to be pointed • Pointing direction • Pointing range • Pointing accuracy • Pointing knowledge • Pointing stability • Slew rate • Exclusion zones	• Entire payload or some payload subset such as antennas or a thermal radiator • Defined relative to what reference • All of the possible pointing directions • Absolute angular control requirement • Knowledge of pointing direction either in real time or after the fact • Maximum rate of change of angular orientation • Reorientation from one pointing direction to another in a specified time • For example, “not within 10° of the Sun”
Other requirements: • Sun pointing • Pointing during thrusting • Communications antenna pointing	• May need for power generation or thermal control • May need for guidance corrections • Toward a ground station or relay satellite

Figure 11.1  gives examples of the inputs and outputs associated with the design and development of the ADCS for a typical spacecraft. Potential trade-offs are identified.

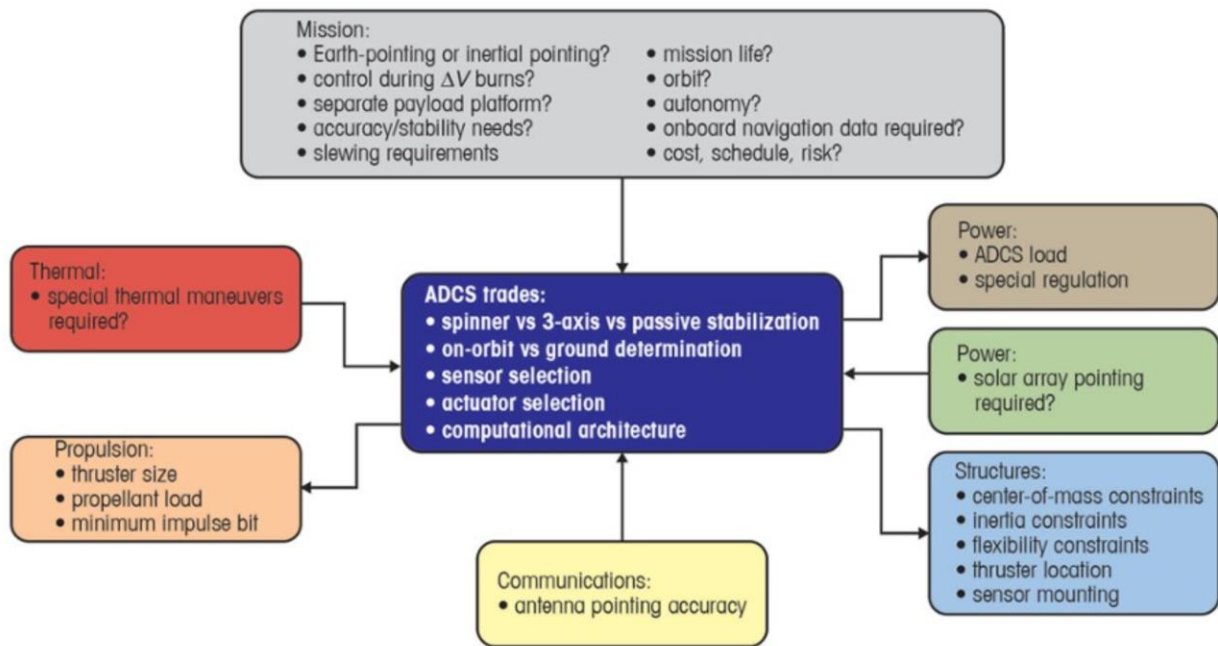


FIGURE 11.1 The Impact of Mission Requirements and Other Subsystems on the Attitude Determination and Control Subsystem (ADCS). The ADCS has strong interactions with most spacecraft subsystems as shown. The direction of arrows shows impacts and potential requirements flow from one subsystem to another. (Adapted from SMAD Fig. 11-2.)

The Attitude Determination and Control Subsystem (ADCS) Challenge

Figure 11.2  portrays a typical closed-loop attitude determination and control subsystem. In a closed-loop control system, the desired attitude is compared with the estimated actual attitude, then an error is determined and fed into the controller. The controller decides on the required action, and commands actuators that modify the pointing direction or spin rate by interacting with the vehicle, which results in a new actual attitude. Sensors gather information on the vehicle's attitude for input to an estimator that applies algorithms to determine the estimated actual attitude, which is then compared with the desired attitude to re-estimate the error.

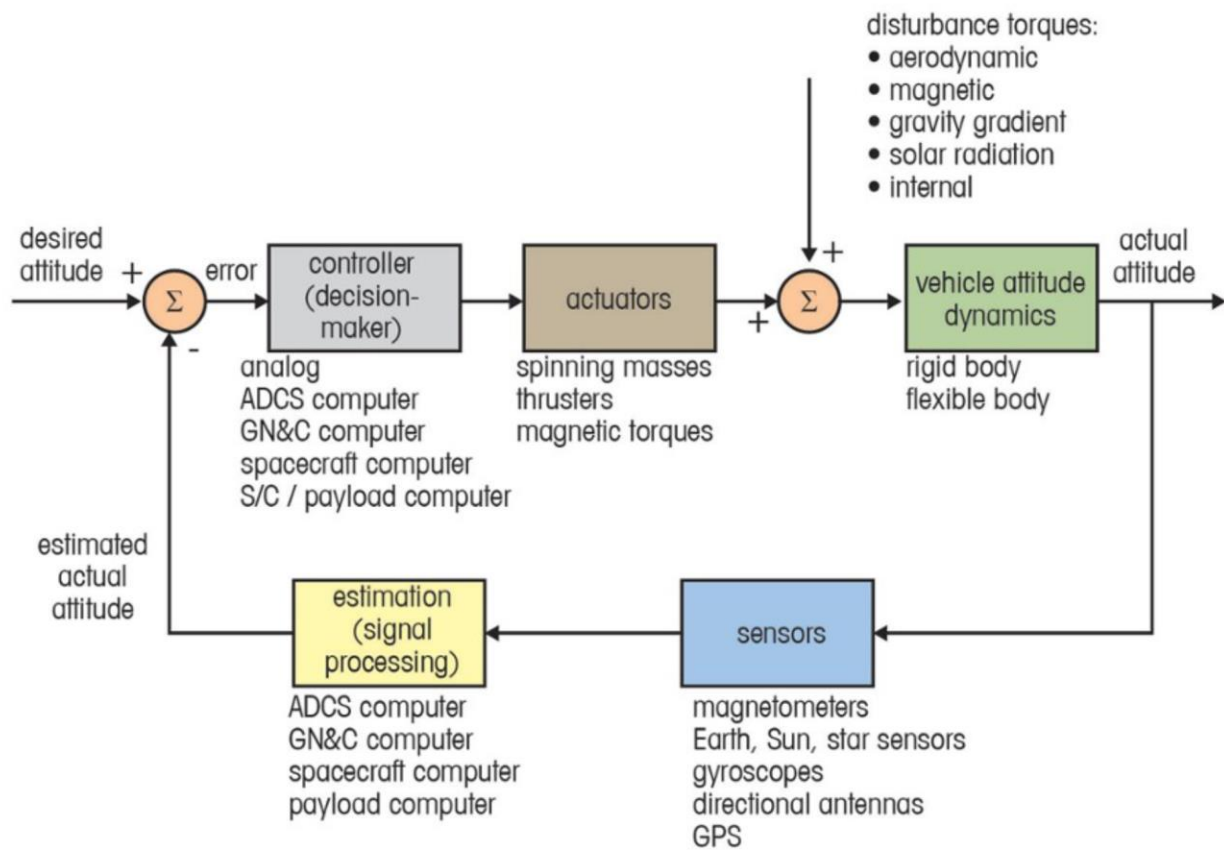


FIGURE 11.2 Closed-loop Attitude Determination and Control. The attitude determination and control subsystem (ADCS) maintains the spacecraft in the desired attitude for the duration of the mission in all control modes.

This section will focus on aspects of this closed-loop control system. The ADCS challenge gets more complex as attitude pointing accuracy and spin rate control requirements become tighter, the number of items to be controlled increases, and the number of control modes increases. In reality the ADCS must deal with all three axes—which are not necessarily independent—in terms of pointing and spin rate control. Disturbance torques tend to rotate the spacecraft in ways that negatively affect the mission, so we have to understand the magnitude and direction of the disturbance torques over time, in order to develop an ADCS capable of maintaining spacecraft attitude.

Attitude Dynamics

For analysis purposes, the rotational motion of the spacecraft can be simplified by applying rigid-body assumptions. This allows us to analyze the most important effects by using Euler's Law of Angular Momentum. The rigid-body simplification may not be appropriate for spacecraft that exhibit significant flexible body effects, such as those with flexible structures or long booms. However, under normal circumstances, non-rigid body effects can be treated effectively as disturbances.

Figure 11.3 defines the coordinate frame used in this section and referred to in other parts of the book. Figure 11.4 provides a slightly different perspective, from the point of view of a generic lifting body. The right-hand rule applies here. Confusion associated with the reference coordinate frame is a source of considerable frustration and rework on the part of the design team, so it is absolutely vital to clearly identify and document the appropriate reference coordinate frame.

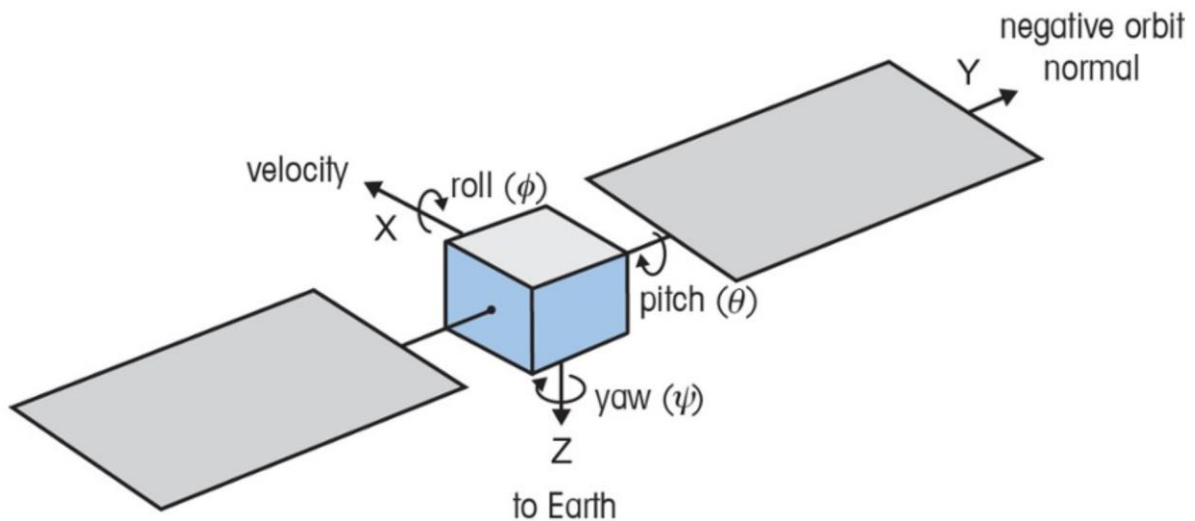


FIGURE 11.3 Example of a Spacecraft-Fixed Coordinate System. To help the reader understand and apply attitude determination and control subsystem (ADCS) principles, we adopt a spacecraft-fixed coordinate frame. The center of this coordinate frame is often specified in spacecraft engineering drawings. The X-axis points in the direction of the velocity vector. The Y-axis points to the negative of the orbit-plane normal, and the Z-axis points to the center of Earth (nadir). Throughout this section, rotations about the X, Y, and Z axes are referred to as roll, pitch, and yaw, respectively. (Adapted from SMAD Fig. 11-1.)

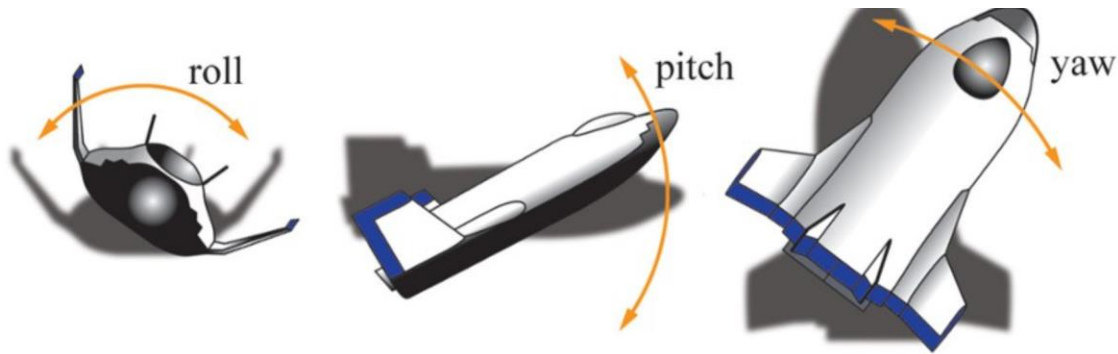


FIGURE 11.4 Another Perspective on Roll, Pitch, and Yaw. The attitude determination and control subsystem (ADCS) design should account for rotations about all three axes. (Adapted from Understanding Space Fig. 12-8.)

The equation of motion for the rotation of a rigid body is expressed as Euler's Law of Angular Momentum, Eq. (11.1) [\[1\]](#), which is a statement of the conservation of angular momentum. This is a vector differential equation. Expanding the left-hand side for a body with constant moments of inertia about the principal axes, we get Eq. (11.2) [\[2\]](#). For a non-spinning spacecraft, we can simplify this to Eq. (11-3) [\[3\]](#). If we apply an external torque to a rotating body, and the torque isn't aligned with the spin vector, the body will precess—rotate at right angles to the applied torque—with a constant angular velocity, as shown in Eq. (11-4) [\[4\]](#).

Eq.11-1

$$\frac{d\mathbf{H}}{dt} = \sum \mathbf{T}_B$$

where

$\mathbf{H}$ = angular momentum

$\mathbf{T}_B$ = torques external to spacecraft body

Eq. 11-2

$$I_x \dot{\omega}_x + (I_z - I_y) \omega_y \omega_z = T_x$$

$$I_y \dot{\omega}_y + (I_x - I_z) \omega_x \omega_z = T_y$$

$$I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y = T_z$$

where

I = moment of inertia

ω = angular velocity

T = the magnitudes of the torque components

For non-spinning spacecraft with small angles and rates:

Eq. 11-3

$$I_x \ddot{\phi} = T_x$$

$$I_y \ddot{\theta} = T_y$$

$$I_z \ddot{\psi} = T_z$$

where

ϕ , θ , and ψ are as defined in [Figure 11.3](#) .

For spinning spacecraft:

Eq. 11-4

$$\omega_p H_{spin} = T$$

where

ω_p = precession rate of the spacecraft's spin vector

H_{spin} = angular momentum of the spacecraft due to spinning

T = net torque external to spacecraft body

Determining Control Modes and Requirements

As we progress through the ADCS design process outlined in [Table 11.1](#), the interactions among the ADCS, the overall mission, and other subsystems will be revealed. As control modes are added to our concept of operations, the ADCS design becomes more challenging and requirements proliferate. [Table 11.3](#) lists typical ADCS control modes. For ADCS performance requirements, see [Table 11.4](#).

Table 11.3 Typical Attitude Determination and Control Subsystem (ADCS) Control Modes. The ADCS may have to function during many different situations or modes. The ADCS requirements may change from one control mode to another, so we must consider all aspects of spacecraft operations. (Adapted from SMAD Table 11-2.)

Mode	Description
Orbit Insertion	Period during and after boost while spacecraft is brought to final orbit. Options include: no spacecraft control; simple spin stabilization (appropriate when orbit insertion propulsion is by solid rocket motor); and full spacecraft control.
Acquisition	Initial determination of attitude and stabilization of vehicle. Also may be used to recover from power upsets or emergencies.
Normal, On-station	Used for the vast majority of the mission. Requirements for this mode should drive system design.
Slew	Reorienting the vehicle when required.
Contingency or Safe	Used in emergencies if regular mode fails or is disabled. May use less power or sacrifice normal operation to meet power or thermal constraints.
Special	Requirements may be different for special targets or time periods, such as eclipses.

Table 11.4 Typical Attitude Determination and Control Subsystem (ADCS) Performance Requirements. These requirements need to be identified for each ADCS control mode. A brief definition of each requirement and examples are provided. (Adapted from SMAD Table 11-3.)

Area	Definition	Examples/Comments
Determination		
Accuracy	How well a vehicle's orientation with respect to an absolute reference is known	0.25°, 3 σ , all axes; may be real-time or post-processed on the ground
Range	Range of angular motion over which accuracy must be met	Any attitude within 30° of nadir
Control		
Accuracy	How well the vehicle attitude can be controlled with respect to a commanded direction	0.25°, 3 σ ; includes determination and control errors, may be taken with respect to an inertial or Earth-fixed reference
Range	Range of angular motion over which control performance must be met	All attitudes, within 50° of nadir, within 20° of Sun
Jitter	A specified angle bound or angular rate limit on short-term, high-frequency motion	0.1° over 1 min, 1°/s, 1 to 20 Hz; usually specified to keep spacecraft motion from blurring sensor data

Drift	A limit on slow, low-frequency vehicle motion. Usually expressed as angle/time.	1°/hr, 5° max. Used when vehicle may drift off target with infrequent resets (especially if actual direction is known).
Settling Time	Specifies allowed time to recover from maneuvers or upsets.	2° max motion, decaying to < 0.1° in 1 min; may be used to limit overshoot, ringing, or nutation

Spacecraft Control Type Selection

There are two general approaches to controlling a spacecraft: passive and active control. With passive control, we take advantage of the interactions between the disturbance torques and the natural responses of the vehicle to keep the satellite stable and oriented in the proper direction, as shown in [Eq. \(11-5\)](#), which are simply [Eq. \(11-2\)](#) with slightly different terminology. With active control, we use feedback control to sense when the vehicle's orientation isn't correct, then apply corrective control torques. See [Eq. \(11-6\)](#).

Passive control is less precise than active, but is simpler and more robust. It's most useful when the vehicle need only be pointed in a certain direction, such as the nadir direction. If we need to slew the vehicle, or components such as instruments, antennas, or solar arrays, we need to employ active control methods.

However, it's not an either-or proposition. Some spacecraft use combinations of active and passive techniques, depending on the control mode appropriate to each mission phase. For instance, during its long-term cruise operations, the Voyager interplanetary spacecraft, launched in 1977, used passive single-spin control, combined with three-axis active stabilization. When it flew by a planet, it used the active control subsystem to slew the spacecraft and hold it steady while imaging the planet and transmitting the information back to Earth.

General Control Strategies

Passive control relies on the dynamics of the spacecraft or the characteristics of the disturbance torques (T_d) to control attitude, as shown in Eq. (11-5) [\[2\]](#).

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Eq. 11-5

$$\begin{aligned}I_x \dot{\omega}_x + (I_z - I_y) \omega_y \omega_z &= T_{d_x} \\I_y \dot{\omega}_y + (I_x - I_z) \omega_x \omega_z &= T_{d_y} \\I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y &= T_{d_z}\end{aligned}$$

Active control uses onboard torque-generating capabilities to achieve the desired attitude and counteract the effects of disturbances. See Eq. (11-6) [\[2\]](#), where the control torques (T_c) have been added to the equations of motion.

Eq. 11-6

$$\begin{aligned}I_x \dot{\omega}_x + (I_z - I_y) \omega_y \omega_z &= T_{c_x} + T_{d_x} \\I_y \dot{\omega}_y + (I_x - I_z) \omega_x \omega_z &= T_{c_y} + T_{d_y} \\I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y &= T_{c_z} + T_{d_z}\end{aligned}$$

Passive Control Methods and Capabilities

Passive control techniques include those shown in Table 11.5 [\[2\]](#). With pure spin stabilization, the entire spacecraft spins about its minimum kinetic energy axis. Gyroscopic stability keeps that axis aligned in the desired direction. Sensors and solar arrays will rotate, and thus need to be designed or chosen appropriately. The spacecraft's structure needs to account for centripetal accelerations, which can be substantial. Changing the orientation of the spin axis is done with actuators, such as thrusters, and can require a good deal of propellant. Examples of pure-spin spacecraft include the Explorer I and Telstar 1 and 2 satellites. Sometimes a spacecraft will be spin stabilized, along with its upper stage, during transfer to the operational orbit.

Table 11.5 Passive Control Methods and Capabilities. Here we present the four basic types of passive control and their essential attributes. (Adapted from SMAD Table 11-4.)

Type	Pointing Options	Attitude Maneuverability	Typical Accuracy	Lifetime Limits
Pure Spin Stabilization	Inertially fixed any direction Repoint with precession maneuvers	High propellant usage to move stiff momentum vector	$\pm 0.1^\circ$ to $\pm 1^\circ$ in 2 axes (proportional to spin rate)	Thruster propellant (if applicable)

Dual-Spin Stabilization	Limited only by articulation on despun platform	Momentum vector same as above Despun platform constrained by its own geometry	Same as above for spin section Despun dictated by payload reference and pointing	Thruster propellant (if applicable) Despin bearings
Gravity-gradient	Earth local vertical only	Very limited	$\pm 5^\circ$ (2 axes)	None
Passive Magnetic	North/south only	Very limited	$\pm 5^\circ$ (2 axes)	None

A variation of this concept is the dual-spin spacecraft, in which a portion of the spacecraft spins, while the rest—usually the central portion—is “despun,” maintaining a fixed attitude in inertial space, allowing for functions like antenna and solar array pointing. The unmanned Defense Satellite Communication System II (DSCS II) spacecraft used dual-spin stabilization, while NASA’s Pioneer 10 and 11 interplanetary probes were single spinners.

As shown in [Eq. \(11-3\)](#) and [Eq. \(11-4\)](#), the responses of spinning and non-spinning spacecraft are inherently different. A non-spinning spacecraft responds to torques with angular accelerations in the same directions as the torques. However, spinning spacecraft, if the applied torque isn’t along the spin axis, respond by precession of the angular momentum vector, at constant angular velocity normal to the torque.

Thus, when the applied torque is removed from a non-spinning spacecraft, the vehicle is left with a constant angular momentum, which may have to be removed by the attitude control system. See the left-hand side of [Figure 11.5](#).

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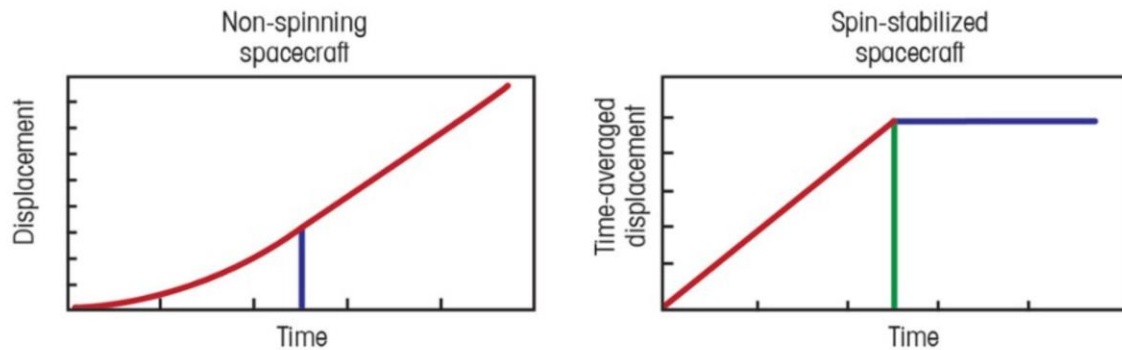


FIGURE 11.5 Displacement of Non-spinning and Spinning Spacecraft. The figure on the left shows a non-spinning satellite's response to the removal of a disturbance torque at the time indicated by the vertical line. The angular acceleration has been nulled out, leaving a residual constant angular velocity. The right-hand figure demonstrates that a spin-stabilized spacecraft has no residual angular velocity. However, it is left with a constant angle error.

On the other hand, when the torque is removed from a spinning spacecraft, the precession-induced angular velocity drops to zero, and the vehicle is left in whatever angular attitude it was when the torque stopped. Thus, it may have an angular error that needs to be nulled.

In addition, spinning spacecraft can experience unwanted oscillations, called nutation and wobble. *Nutation* is an oscillation of the angular momentum vector, caused when the spin axis isn't aligned with a principal axis. It can be caused by thruster firings, structural flexing, and fuel sloshing. We can use nutation dampers to eliminate nutation and produce "clean" spin. On the other hand, *wobble* occurs when the spin axis is aligned with a principal axis, but that axis isn't aligned with the body reference axis—caused by imbalance.

Gravity-gradient stabilization is a naturally-occurring phenomenon that we can use to keep a spacecraft pointing toward nadir. It makes use of the vehicle's distribution of mass and the r -squared nature of gravitational force and the $1/r$ nature of centripetal force. [Figure 11.6](#) and inequalities [Eq. \(11-7\)](#) illustrate the physics by simplifying the orbiting body in the form of a symmetrical dumbbell that is not aligned with the vector from Earth's center. Although the two lobes of the dumbbell have equal mass, the net force on the lower lobe is greater than that on the upper lobe, creating a torque that rotates the lower lobe downward. Because the distance between the lobes is a very tiny fraction of the total radial distance (R) the torque is very small—typically on the order of 10^{-5} N·m. Thus, the restoring accelerations are very low, and the ability to null disturbance torques is modest. As a result, the vehicle pointing accuracy is also modest (about 5°). However, gravity-gradient stabilization requires no power or consumables, and is thus "free." This technique will only stabilize the spacecraft's pitch and roll axes—not the yaw axis. [Eq. \(11-8\)](#) gives the maximum torque magnitude for a given pitch/roll angular offset from nadir. We can use this equation for estimating gravity gradient effects on the spacecraft.

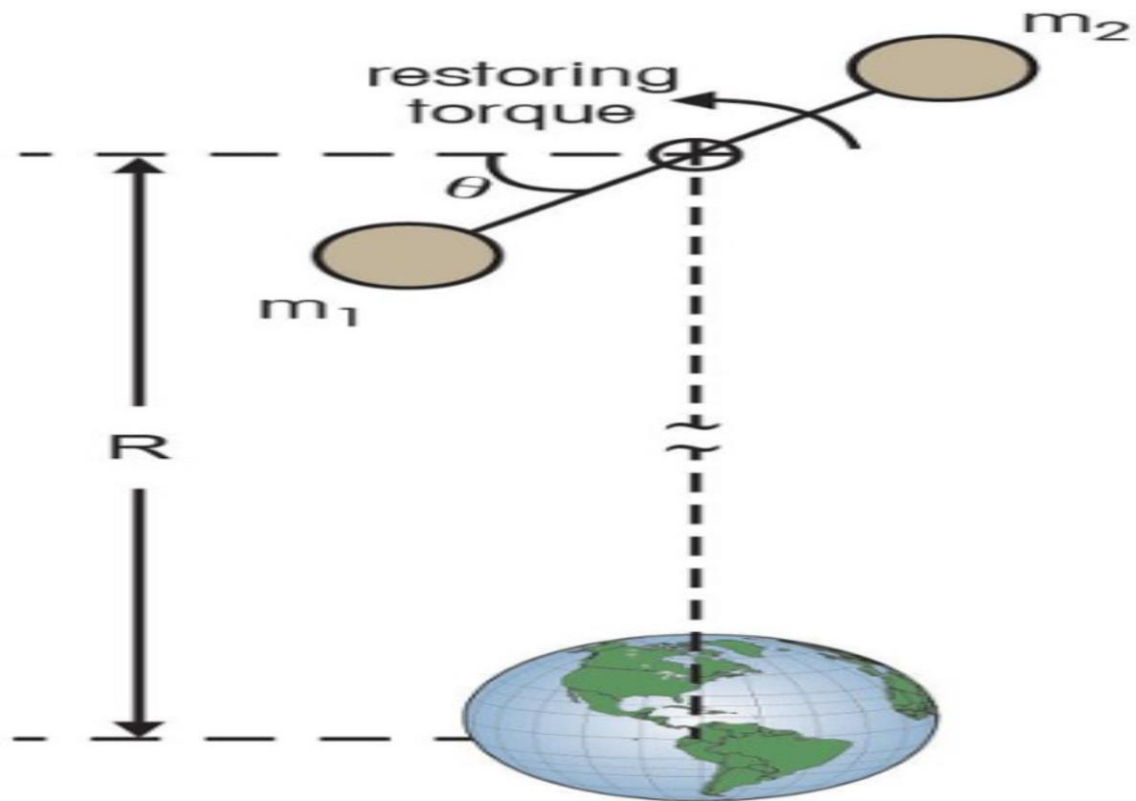


FIGURE 11.6 Illustration of the Physics of Gravity Gradient Stabilization. Any body with distributed mass will exhibit these restoring torques, which are generally quite small, but are “free” and don’t use consumables or power. Sometimes, long booms are deployed from the spacecraft, to magnify the effect. (Adapted from Understanding Space, Fig. 12-17.)

Depending on the vehicle’s mass distribution (moments of inertia), gravity-gradient stabilization may or may not be stable. See [Eq. \(11-9\)](#) 

Our moon always presents the same side to Earth because it's gravity-gradient stabilized, or tidally locked in its orbit.

Referring to [Figure 11.6](#) ,

Eq. 11-7

$$\text{at position of } m_1 : F_g = m_1 \omega_o^2 R > 0$$

$$\text{at center of mass: } F_g - m \omega_o^2 R = 0$$

$$\text{at position of } m_2 : F_g - m_2 \omega_o^2 R < 0$$

where

m = spacecraft's total mass

m_1 and m_2 = masses of the lobes

F_g = magnitude of total gravity force on spacecraft

F_{g1} and F_{g2} = magnitude of gravity forces on each lobe ($F_{g1} > F_{g2}$ as shown in [Figure 11.6](#) )

ω_o = magnitude of satellite's orbital angular velocity

Eq. 11-8

$$T_g = \left(\frac{3\mu}{2R^3} \right) |I_z - I_y| \sin(2\theta)$$

where

T_g = magnitude of maximum torque due to gravity gradient

μ = Earth's gravitational constant, $3.986 \times 10^{14} \text{ m}^3/\text{s}^2$

R = radial distance to the center of Earth, in m
 θ = total pitch or roll angle offset from nadir, in radians

The conditions for gravity-gradient stabilization to always be stable are given as:

Eq. 11-9

$$I_{pitch} > I_{roll} > I_{yaw}$$

where

$$I_{pitch} = I_Y$$

$$I_{roll} = I_X$$

$$I_{yaw} = I_Z$$

On the other hand, it will only be stable sometimes if the following inequalities are met:

Eq. 11-10

$$I_{roll} > I_{yaw} > I_{pitch}$$

Passive magnetic stabilization is useful in relatively low, near-equatorial Earth orbits. This technique uses permanent magnets to force the spacecraft to align with Earth's magnetic field. Because this approach uses no moving parts and doesn't require propellant, it is very reliable and allows for long service lifetimes.

Active Control Methods and Capabilities

Active control schemes use either angular-momentum devices or thrusters to provide control torques.

Table 11.6  describes three types of the former and one of the latter.

Table 11.6 Active Control Methods and Capabilities. Here we present the four basic types of active control and their essential attributes. The accuracies shown are for bus-pointing during a nominal, Earth-pointing mode only, and do not necessarily represent payload-pointing accuracies, or accuracies during maneuvers. (Adapted from SMAD Table 11-4.)

Type	Pointing Options	Attitude Maneuverability	Typical Accuracy	Lifetime Limits
Bias Momentum (1 wheel)	Best suited for local vertical pointing	Momentum vector of the bias wheel prefers to stay normal to orbit plane, constraining yaw maneuver	$\pm 0.1^\circ$ to $\pm 1^\circ$	Propellant (if applicable) Life of sensor and wheel bearings
Zero Momentum (thruster only)	No constraints	No constraints; high rates possible	$\pm 0.1^\circ$ to $\pm 5^\circ$	Propellant
Zero Momentum (3–4 reaction wheels)	No constraints	No constraints	$\pm 0.001^\circ$ to $\pm 1^\circ$	Propellant (if applicable) Life of sensor and wheel bearings

Zero Momentum CMG	No constraints	No constraints; high rates possible	$\pm 0.001^\circ$ to $\pm 1^\circ$	Propellant (if applicable) Life of sensor and wheel bearings
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The choice of control method depends on the accuracy desired, the available mass and power, and the expected lifetime of the spacecraft—as well as cost. Below is a description of each of the methods shown in the table:

- Bias momentum systems
 - ~ Momentum wheels usually rely on a single wheel with a large, controllable momentum to provide overall stiffness and some variability to reduce disturbance torques by adjusting the wheel's speed
- Zero-bias systems
 - ~ Reaction wheels that rely on three or more wheels, normally with little or no initial momentum ([Figure 11.7](#))
 - ~ Each wheel spins independently to rotate the spacecraft and absorb disturbance torques
- Control-moment gyroscopes (CMGs)
 - ~ CMGs rely on three or more wheels, each with a large, fixed momentum
 - ~ The wheels are mounted on gimbals; rotating the wheels about their gimbals changes the spacecraft orientation ([Figure 11.8](#))
- Thrusters
 - ~ Use either cold-gas or hot-gas jets, oriented in pairs to provide a nearly pure couple (moment)

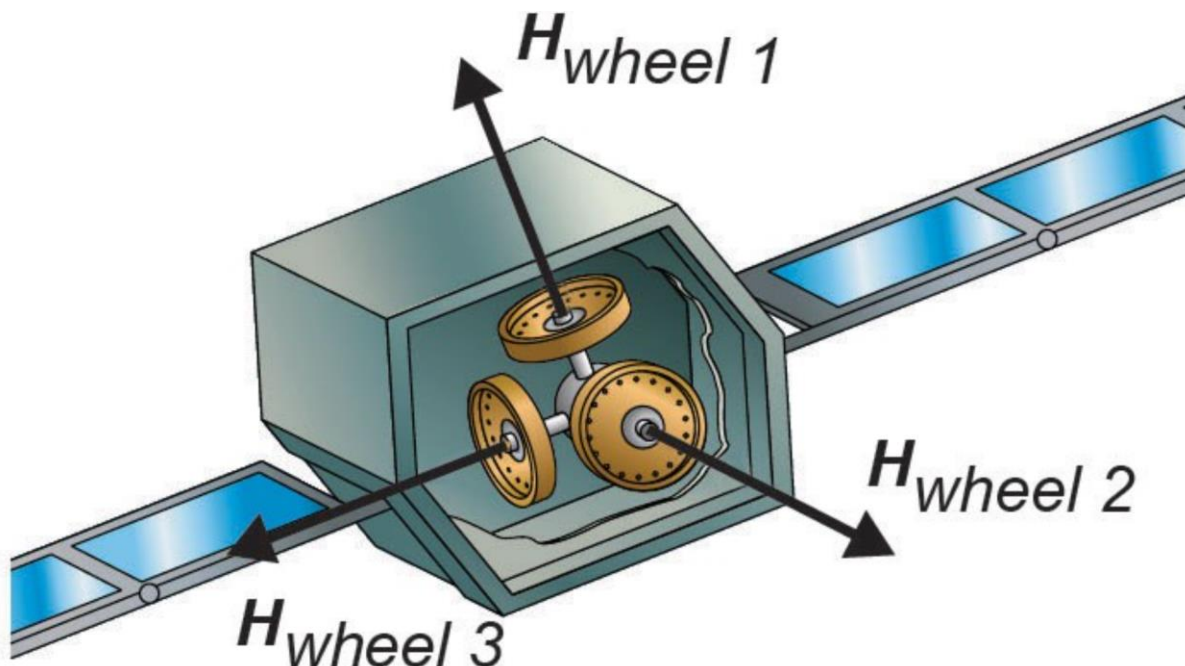


FIGURE 11.7 Reaction Wheels. Here we show a three-wheel, zero-bias system. (Adapted from Understanding Space Fig. 12-33.)

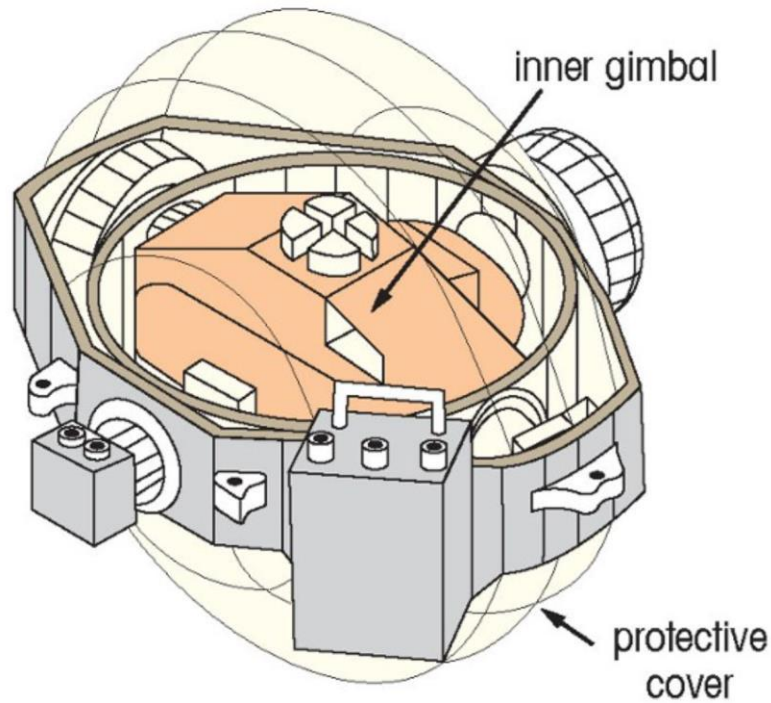


FIGURE 11.8 Control Moment Gyro (CMG) Package. The International Space Station uses 4 CMGs, each with a mass of ~250 kg and spinning at 6600 rpm.

Simplified Equations for Wheel Sizing

In [Table 11.7](#), we present equations for the preliminary sizing of reaction wheels and momentum wheels. For reaction wheels, we need to design for both worst-case disturbance torques and for the desired vehicle slew rate. As the wheel spins up as a result of providing the desired torques, it stores angular momentum, and we must design the wheel's moment of inertia and maximum speed to accommodate this. As shown in the table, we can estimate the angular momentum storage requirement by considering the worst-case cyclic disturbance torque, integrated over half an orbit. (If we're trying to null out the gravity-gradient effect on the satellite, then we use one quarter of an orbit.)

Recall that a momentum wheels has a large, controllable angular momentum to provide gyroscopic stiffness. We estimate the wheel's angular momentum by considering the worst-case disturbance torque and the required pointing accuracy, as shown in the equation in the bottom of [Table 11.7](#).

Table 11.7 Simplified Equations for Wheel Sizing. Note the differences in sizing relations between reaction wheels and momentum wheels. Equations give magnitudes only. (Adapted from SMAD Table 11-12)

Parameter	Simplified Equations
Torque from reaction wheel for disturbance rejection	Reaction wheel torque (M_{RW}) must equal worst-case anticipated disturbance torque (T_D) plus some margin: $M_{RW} = T_D (1 + \text{Margin Factor})$
Slew torque for reaction wheels	For maximum acceleration slew (1/2 distance in 1/2 time): $M = 4I\theta / t^2$ where I is moment of inertia, θ is slew angle, and t is time for the slew maneuver

Momentum storage in a reaction wheel	One approach to estimating momentum capacity (H) is to integrate the worst-case cyclic disturbance torque (T_D) over half an orbit. If the disturbance is magnetic, the maximum disturbance is accumulated in one quarter of an orbit. A simplified expression for such a sinusoidal disturbance is: $H = 0.707 T_D [\text{Orbital Period} / 4]$ where 0.707 is the rms average of a sinusoidal function
Momentum storage in a momentum wheel	Roll and yaw accuracy depend on the wheel's momentum and the external disturbance torque (M_D). A simplified expression for the required momentum storage (H) is: $H = T_D [\text{Orbital Period} / (4\theta_a)]$ where θ_a is the allowable motion

Active Control: Thrusters

Thrusters have a variety of uses on spacecraft, from nulling disturbance torques, to slewing, to providing the moments needed to hold the vehicle steady during momentum dumping of reaction wheels or momentum wheels. [Table 11.8](#) gives simplified relations for sizing thrusters.

Table 11.8 Approximate Relations for Thruster Sizing. Note the variety of thrusting situations presented. We base thruster sizing on the most demanding of the applicable situations. Equations give magnitudes only. (Adapted from SMAD Table 11-13.)

Parameter	Simplified Equations
Thruster force for external disturbances	$F = T_D / L$ where F is thruster force, T_D is worst-case disturbance torque, and L is thruster moment arm
Thruster force to meet slew rates for a zero-momentum (reaction wheel or thruster-only) system	Determine the highest slew rate required in the mission profile. Develop a profile that accelerates the vehicle to that rate, coasts, then decelerates. The thruster force can be calculated from the acceleration value using the following relationship: $F = I\ddot{\theta} / L$ where F is thruster force, I is moment of inertia, $\ddot{\theta}$ is angular acceleration, and L is the moment arm
Thruster force to slew a momentum-bias vehicle	$F = H \dot{\theta}_{slew} / Ld$ where F is the average thruster force, L is the moment arm, d is the thruster duty cycle (fraction of spin period), H is the angular momentum of the spacecraft ($I\omega$), and $\dot{\theta}_{slew}$ is the slew rate

Thruster force for momentum dumping	$F = H/Lt$ where F is thruster force, H is the stored angular momentum (from wheel capacity or disturbance torque $\times$ time), L is the moment arm, and t is the burn time
Thruster pulse life	Develop detailed maneuver profile from mission sequence of events and determine pulse number and length for each segment

Thrusters are employed in pairs to provide nearly pure moments (couples) about the vehicle axes. We approximate the response of the vehicle to thruster-induced moments by using the linearized differential equations of motion, [Eq. \(11-3\)](#), which are adapted for controller moments in [Eq. \(11-11\)](#). The M_c are the thruster-generated control moments (torques). More precise design must account for the non-linear effects, such as deadband (limit cycle) and hysteresis. [Figure 11.9](#) demonstrates an attitude control scheme using thrusters. We design switching curves in the control system logic to determine when the thrusters will fire to control the limit cycle. (See the subsection below: “Defining Controller Algorithms.”)

Eq. 11-11

$$I_x \ddot{\phi} = M_{cx}$$

$$I_y \ddot{\theta} = M_{cy}$$

$$I_z \ddot{\psi} = M_{cz}$$

Active Control: Magnetic Torquers

Magnetic torquers (also called magnetorquers) can be useful in Earth orbits ranging from low-Earth to geosynchronous. When used in an active control mode, this type of effector passes electrical current through electromagnetic coils to produce a magnetic field that interacts with Earth’s own magnetic field, to produce a torque on the spacecraft. By switching the electromagnetic coils on or off at the appropriate times, we can control the vehicle’s attitude. For nadir-pointing satellites, they can generate control moments in both yaw and roll. They can also help “capture” a gravity gradient attitude. And they can desaturate reaction wheels or momentum wheels. For spinning satellites, they can help provide spin-rate control and can precess the vehicle.

As mentioned earlier, magnetic torquers can also be used passively, if we use permanent magnets, rather than electromagnets.

The moment generated by a magnetic torquer is expressed by [Eq. \(11-12\)](#). See also [Figure 11.9](#).

Eq. 11-12

$$M = D \times B$$

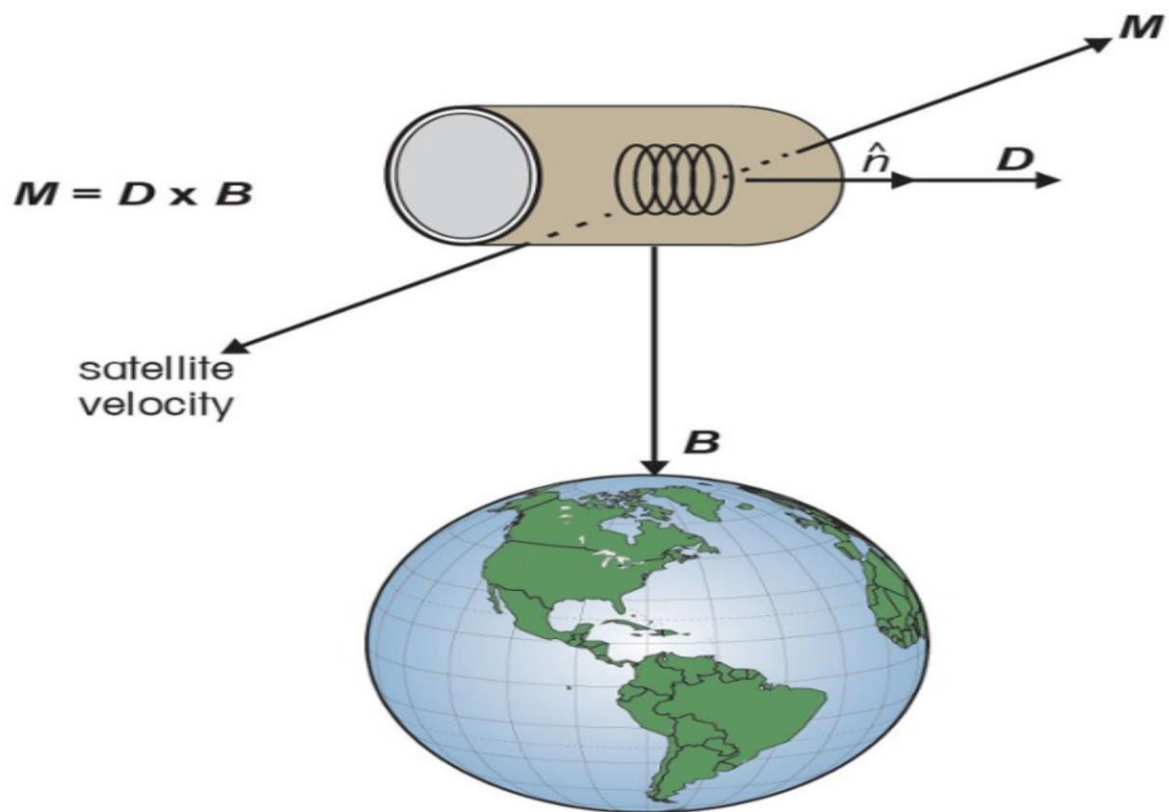


FIGURE 11.9 Magnetic Torquer. The moment generated by the torquer is proportional to, and normal to, the dipole moment of the electromagnet and Earth's magnetic field. The moments are generally small, but can very useful for small spacecraft. In this figure, the torquer's dipole moment vector (D) is pointing normal to the orbit plane. The resulting control moment is along the satellite's roll axis.

In this equation, $\mathbf{D}$ is the magnetic dipole moment of the electromagnet, and $\mathbf{B}$ is Earth's magnetic field, in tesla. $\mathbf{D}$ is equal to the product of the number of turns in the coil, the current, and the coil's area, as shown in Eq. (11-13) [↗](#). The units of $\mathbf{D}$ are amp·turns·m².

Eq. 11-13

$$\mathbf{D} = NIA\hat{n}$$

where

N = number of turns

I = current (not the moment of inertia)

A = coil area

$\hat{n}$ = direction vector, which is normal to the plane of the coil

Because magnetorquers have no moving parts and don't require propellant, they are very reliable and have long service lifetimes. However, they generate relatively low torques, so aren't suited to precise control of large spacecraft, unless the current is very high and the coils are very large. Magnetic torquers also aren't of use high-Earth orbit or deep space missions, or in orbit around bodies such as the Moon or Mars, which have very weak magnetic fields.

Quantifying the Disturbance Environment

Now we address the problem of estimating the disturbance torques on the spacecraft. Those disturbances are due to both external and internal sources. Causes of external torques include gravity gradient, solar radiation pressure, Earth's magnetic field interacting with the vehicle's electromagnetic field, and aerodynamic forces. See [Table 11.9](#) . [Figure 11.10](#) presents representative curves of external torque values for Earth-orbiting satellites. [Table 11.10](#) gives approximate formulas for calculating worst-case external disturbance torques.

Table 11.9 External Disturbance Sources and their Characteristics. Some disturbances are cyclic over the course of an orbit, while others are secular. (Adapted from SMAD Table 11-9A.)

Disturbance	Type	Influenced Primarily by
Gravity-gradient	Constant torque for Earth-oriented vehicle in circular orbit, cyclic for inertially oriented vehicle	• Spacecraft moments of inertia • Orbit altitude
Solar Radiation	Cyclic torque on Earth-oriented vehicle, constant for solar-oriented vehicle or platform	• Spacecraft surface reflectivity • Spacecraft geometry and c_m location
Magnetic Field	Cyclic	• Orbit altitude • Residual spacecraft magnetic dipole • Orbit inclination
Aerodynamic	Constant for Earth-oriented vehicle in circular orbit, variable for inertially oriented vehicle	• Orbit altitude • Spacecraft geometry and c_m location

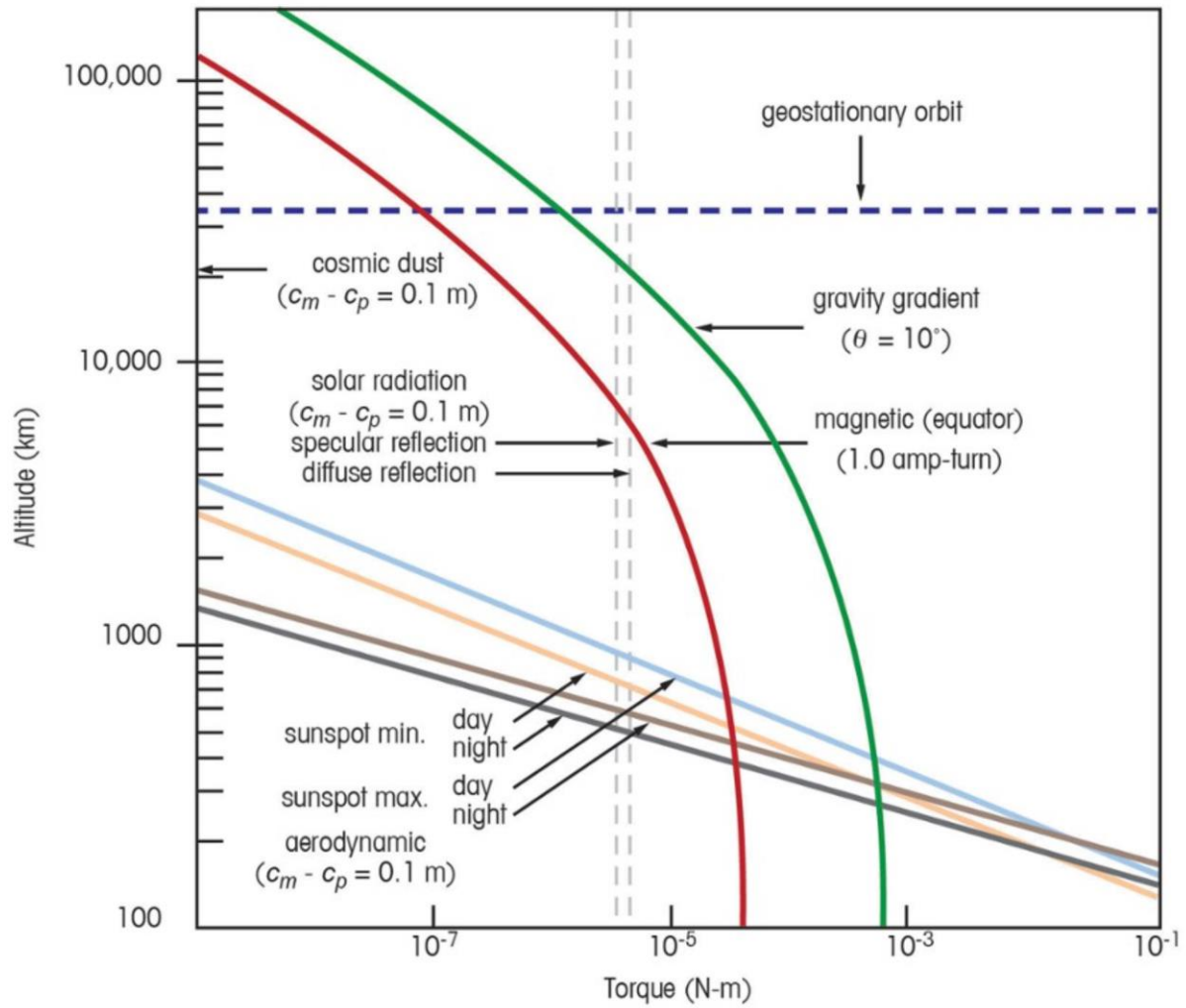


FIGURE 11.10 External Disturbance Torques for Earth-orbiting Satellites. The curves show approximate torque values, as functions of orbit altitude. (c_m = center of mass; c_p = center of pressure.) (Figure from Spacecraft Attitude Dynamics, by Hughes, Fig 8.15.)

Table 11.10 Estimating Magnitudes of the Worst-case External Torques. These approximate equations provide useful estimates for use in preliminary sizing of control devices, such as reaction wheels, momentum wheels, thrusters, and magnetic torquers. (Adapted from SMAD Table 11-9A.)

Disturbance	Formula
Gravity Gradient	$T_g = 3\mu I_z - I_y \sin(2\theta) / 2R^3$ where T_g is the max gravity torque; μ is Earth's gravity constant ($3.986 \times 10^{14} \text{ m}^3/\text{s}^2$); R is orbit radius (m), θ is the maximum deviation of the Z-axis from local vertical in radians, and I_z and I_y are moments of inertia about z and y (or x, if smaller) axes in $\text{kg}\cdot\text{m}^2$.
Solar Radiation	Solar radiation pressure depends strongly on the type of surface being illuminated. A surface can be transparent, absorbent, or reflective, and most surfaces are a combination of the three. Reflectors are classed as diffuse or specular. In general, we consider solar arrays to be absorbers and treat the spacecraft body as a reflector. The worst-case solar radiation torque is $T_{sp} = F(c_{ps} - c_m) \text{ where } F = (1/c)F_s A_s (1 + q) \cos i$ and F_s is the solar constant ($1,367 \text{ W}/\text{m}^2$), c is the speed of light ($3 \times 10^8 \text{ m/s}$), A_s is the surface area, c_{ps} is the location of the center of solar pressure, c_m is the center of mass, q is the reflectance factor (ranging from 0 to 1; for simplicity, when doing rough estimates, we use 0.6), and i is the angle of incidence of the Sun.
Magnetic Field	$T_m = DB$ where T_m is the magnetic torque on the spacecraft; D is the residual dipole of the vehicle in $\text{amp}\cdot\text{turn}\cdot\text{m}^2$ ($\text{A}\cdot\text{m}^2$), and B is Earth's magnetic field in tesla. B can be approximated as $2M/R^3$ at the poles or half that at the equator. M is Earth's magnetic moment, $7.96 \times 10^{15} \text{ tesla}\cdot\text{m}^3$, and R is the distance from the center of Earth to the spacecraft in m.

Aerodynamic	$T_a = F (c_{pa} - c_m)$ where $F = 0.5 [\rho C_d AV^2]$; F is the force; C_d the drag coefficient (usually between 2 and 2.5); ρ the atmospheric density, which varies significantly with altitude and solar activity; A is the surface area; V is the spacecraft velocity; c_{pa} is the center of aerodynamic pressure; and c_m is the center of mass.
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As shown in [Table 11.11](#) , internal disturbances result from phenomena such as unbalanced thruster torques caused by the following: uncertainty in the location of the vehicle's center of mass; misalignment of thrusters; and unequal (mismatched) thrust outputs of paired thrusters. For spinning spacecraft, these can lead to wobble and nutation. Other internal torques come from rotating hardware, such as solar arrays and antennas, manipulator arms, fans, and pumps. Fuel sloshing both creates disturbance torques and dissipates rotational kinetic energy of a spinning satellite. Flexing of large structures, such as solar panels, trusses, and booms, also creates internal disturbances and absorbs kinetic energy. Similarly, structural vibrations can be caused by thermal shocks such as those that occur when parts of a satellite suddenly heat up or cool down, as when entering or leaving eclipse.

Table 11.11 Internal Disturbance Sources and their Characteristics. Even with very precise design of the spacecraft, these sources are difficult to eliminate totally. (Adapted from SMAD Table 11-10.)

Disturbances Source	Effect on Vehicle	Typical Values
Uncertainty in Center of Mass (c_m)	Unbalanced torques during firing of coupled thrusters Unwanted torques during translation thrusting	1 to 3 cm
Thruster Misalignment	Same as c_m uncertainty	0.1 to 0.5°

Mismatch of Thruster Outputs	Similar to c_m uncertainty	$\pm 5\%$
Rotating Machinery (pumps, articulated solar arrays)	Torques that perturb both stability and accuracy	Depends on spacecraft design; may be compensated by counter-rotating elements
Liquid Sloshing	Torques due to fluid motion and variation in center of mass location	Depends on specific design; may be controlled by bladders or baffles
Dynamics of Flexible Bodies	Oscillatory resonance at bending frequencies, limiting control bandwidth	Depends on spacecraft structure
Thermal Shocks on Flexible Appendages	Attitude disturbances when entering or leaving eclipse	Depends on spacecraft structure. Worst for gravity-gradient systems with long inertia booms.

Selecting and Sizing ADCS Hardware

Having characterized the various types of actuators and estimated the disturbance torques, we are ready to do preliminary sizing of the actuators. [Table 11.12](#) gives us rough estimates of the mass and electrical power required for four types of actuators. This process is iterative with the rest of the vehicle design.

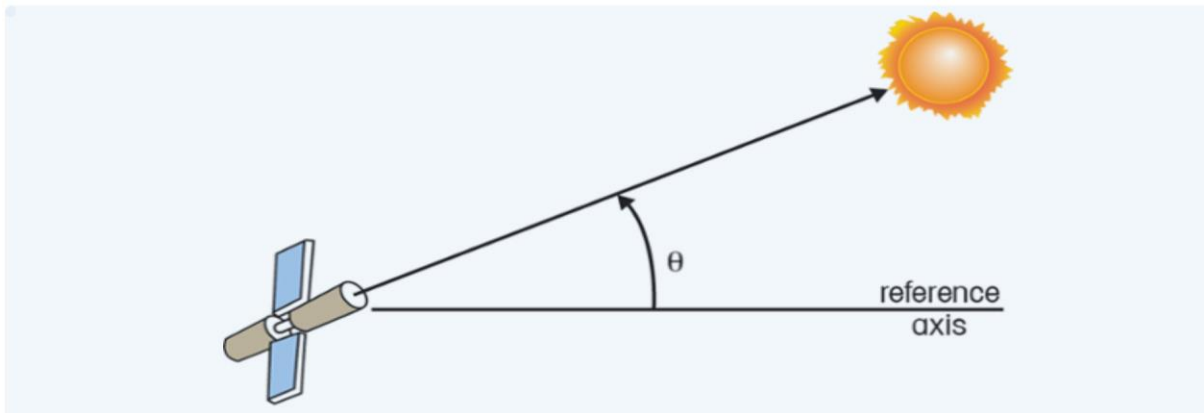
Attitude Determination Sensors

To implement feedback control of the satellite's attitude, we must measure the attitude and attitude rates. Refer to the lower half of [Figure 11.2](#). Attitude sensors in use today fall into the following categories:

- Sun sensors, horizon sensors, and star sensors
- Gyroscopes
- Magnetometers
- Global Positioning System (GPS)

Sensor measurements are typically fed into an estimator that “filters” multiple inputs to determine a stable prediction of current attitude and attitude rates.

Figure 11.11 [🔗](#) demonstrates the operation of *sun sensors* and *star sensors*. In simple terms, they determine the angle between a vehicle reference axis and the vector to the sun or another star. These sensors are most useful for inertially-pointing spacecraft.



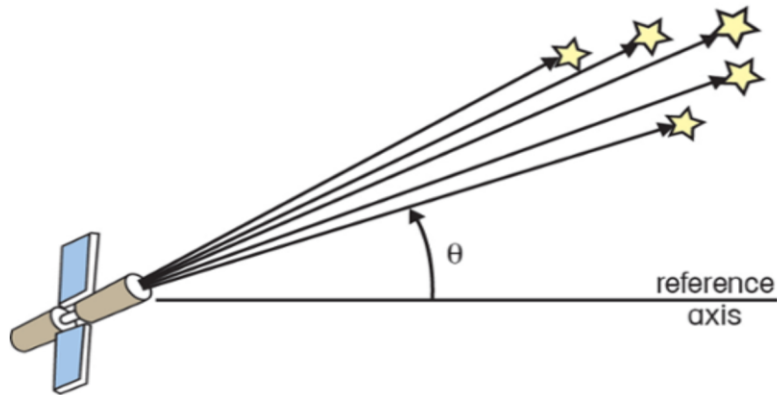


FIGURE 11.11 Sun Sensors and Star Sensors. (Figures from Understanding Space, Figs. 12-19 and 12-20.)

Table 11.12 Sizing of Typical Actuators. Use this table for rough estimates of actuator mass and required electrical power for various types of actuators. (Adapted from SMAD Table 11-11.)

Actuator	Typical Performance Range	Mass (kg)	Power (W)
Thrusters	• Microsats: 100s of N	< 5 kg	(Pulsed plasma)
	• Mid-size (LEO): 10s–100s of N	Variable	N/A (hydrazine)
	• Large sats (GEO): 100s–1000s of N	Variable	N/A (hydrazine)
Reaction and Momentum Wheels	• Nanosats: 1–20 mN · m · s; torque 0.6 mN · m	0.6 kg (3-wheel pkg)	2–6 W
	• Small sats (LEO): 1–12 N · m · s; torque 0.050–0.150 N · m	3–5 kg per wheel	10 W
	• Mid-size (LEO): 30 n · M · s; torque 0.4 N · m	<16.5kg/wheel	35 W

	• Large sats (GEO): 100–400 N · m · s; torque 0.1–1 N · m	<16.5 kg/wheel	35–100 W
Control Moment Gyros (CMG)	• Microsats: torque 0.4 N · m	1.2 kg (4-gyro cluster)	1.6 W
	• Small sats (LEO): torque 25–75 N · m	10–30 kg	35–95 W
	• Mid-size (LEO): torque 75–100s of N · m	10–30 kg	35–95 W
	• Large sats (GEO): torque 100s of N · m	> 30 kg	90–150 W
Magnetic Torquers	• Microsats: 0.1–1.0 A · m ²	0.003–0.06 kg	0.15–0.3 W
	• Small sats (LEO): 30 A · m ²	0.6 kg	4 W
	• Mid-size (LEO): 400 A · m ²	8 kg	11.5 W
	• Large sats (LEO): 3500 A · m ²	38 kg	10–20 W

For Earth-pointing spacecraft, *horizon sensors* measure the orientation of the vehicle’s reference axes, relative to Earth’s horizon. By their nature, horizon sensors allow determination of the vehicle’s pitch and roll angles, but not yaw. Another sensor type must be used to measure yaw.

Magnetometers measure the strength and direction of Earth’s magnetic field vector. [Figure 11.12](#) illustrates one type of magnetometer design used on spacecraft, the “search coil” design.

We can use [Table 11.13](#) to select the appropriate attitude-determination sensors for our spacecraft, based on the required accuracy. The table also gives the mass and required electrical power for each type, for sizing.

Table 11.13 Selecting Attitude Sensors. This table presents representative performance levels for various sensor types. It also gives their sizing information. (Adapted from SMAD Table 11-14.)

Sensor	Typical Performance Range	Mass (kg)	Power (W)
Inertial measurement unit (IMUs)	Microsats: 0.5° to 1°	0.06	< 0.2
	Mid-size (LEO): 0.06°	5	30
	Large sats (GEO): 0.04°	5	30
Sun sensors	Microsats: 0.5°	0.005	0.05
	Mid-size (LEO): 1°	0.3	0.01 to 0.1
	Large sats (GEO): 0.6°	1	< 3
Star sensors (scanners and mappers)	Microsats: 0.002° to 0.011°	0.35	< 0.5
	Mid-size (LEO): 0.0008° to 0.003°	< 3	< 10
	Large sats (GEO): 0.0008° to 0.003°	< 5	18
Horizon sensors	Microsats: 0.2°	0.085	0.4
	Mid-size (LEO): 0.2°	< 4	4
	Large sats (GEO): 0.05°	7.5	5
Magnetometer	Microsats: 1°	0.015	0.4
	Mid-size (LEO): 0.5° to 3°	0.3 to 1.2	<1
Global Positioning System (GPS)	Microsats: < 1°	0.03 plus antennas	< 1
	Mid-size (LEO): 1.2° to 1.6°	1	5.5
	Large sats (GEO): 0.3°	4.5 incl. antennas	10

Sometimes tight accuracy requirements dictate the use of multiple attitude sensors to produce a single estimate of attitude and attitude rates. We then use an estimator algorithm to combine the measurements to provide a more reliable estimate of the “state,” for use by the controller. The Kalman filter is one such approach. We further discuss Kalman filters in the subsection “Define Controller Algorithms.”

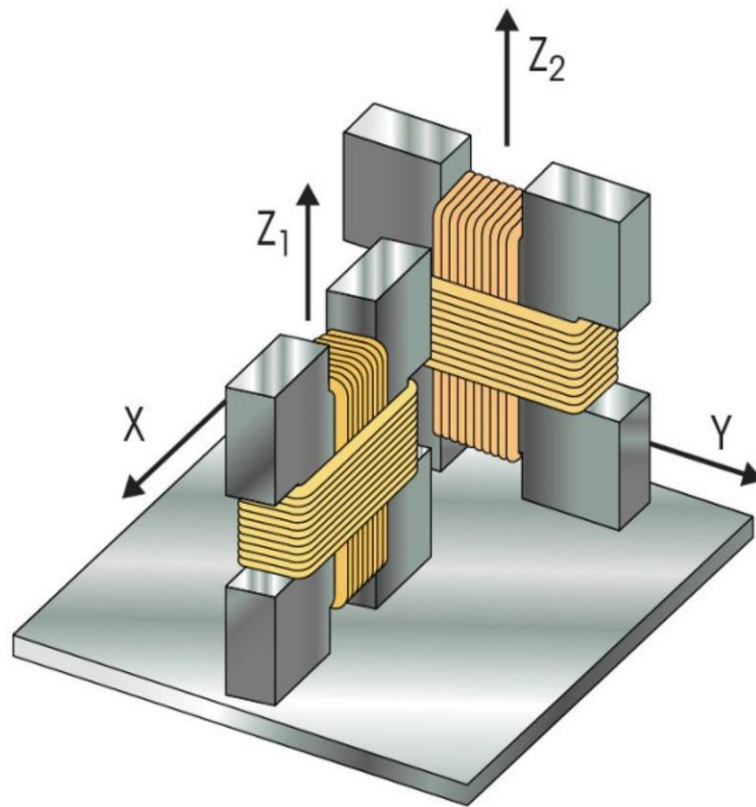


FIGURE 11.12 Magnetometer. This device measures the magnitude and direction of Earth's magnetic field, for comparison with a highly precise model of the magnetic field, to yield vehicle attitude. This example is a search coil design. Each coil is sensitive to the magnetic field strength along its winding axis. In the configuration shown, one coil determines the component along the vehicle's x direction and another coil, the y direction. Two coils measure along the z direction. (Figure from Understanding Space, Fig. 12-22.)

Defining Estimation and Control Algorithms

To generate the desired vehicle response to control inputs, a software-implemented controller—or compensator—determines the appropriate commands for the actuators. In [Figure 11.2](#), this is depicted as the box labeled “controller (decision-maker).”

The controller takes as input the error in the vehicle’s attitude state (attitude components and attitude rates)—that is, the difference between the desired state and the estimate of the actual state. It then determines the input signals for the actuators, based on the desired response of the vehicle to the error. For design purposes, the desired response is often expressed in terms of speed of response, amount of overshoot (or undershoot), and steady-state error for various types of vehicle commands, such as a step change in vehicle attitude.

Various mathematical schemes are available for ensuring these desired responses. When the actuators provide a smooth, continuous range of torque values (such as for momentum wheels and control moment gyros) typical control schemes include proportional, derivative, and integral compensation. Usually, these three types are combined into proportional-plus-derivative (PD), proportional plus integral (PI), or proportional-plus-integral-and-derivative (PID). In their practical implementations, these schemes employ a set of gains that can be altered to achieve the desired response. [Figure 11.13](#) shows a PID compensator for controlling a single axis of rotation. (When using 3-axis control, we often treat each axis independently.) The input is the attitude error ($e(t)$) and the output ($u(t)$) is the control signal that goes to the actuators. The designer selects the control gains, K_p , K_D , and K_I to achieve the desired transient response (overshoot/undershoot and speed of response) and steady-state error. [Eq. \(11.14\)](#) is the equation for $u_{PID}(t)$:

Eq. 11-14

$$U_{PID} = K_p + K_D \frac{de}{dt} + K_I \int e(t) dt$$

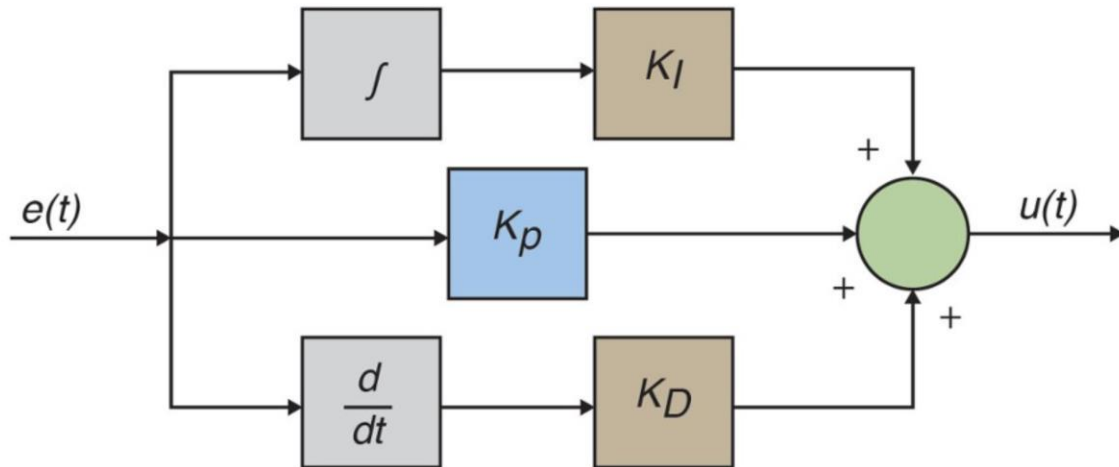


FIGURE 11.13 Block Diagram of a Proportional-Plus-Integral-and-Derivative (PID) Compensator for Control of a Single Axis of Rotation. The input is the attitude error ($e(t)$). The output is the control signal that is fed to the actuators. The output is a linear combination of the error at the current time, the integral of past errors, and the error rate at the current time. The controller gains are: K_p (proportional), K_I (integral), and K_D (derivative).

When the actuators are thrusters, the compensator design is somewhat different. Thrusters are normally not throttled, so the control moments they generate are either “on” or “off.” The compensator logic then becomes a set of “switching curves” that determines when either the positive or negative thruster pairs should be fired. This is the classic “bang-off-bang” control problem. [Figure 11.14](#)  is an example of switching curves for a satellite.

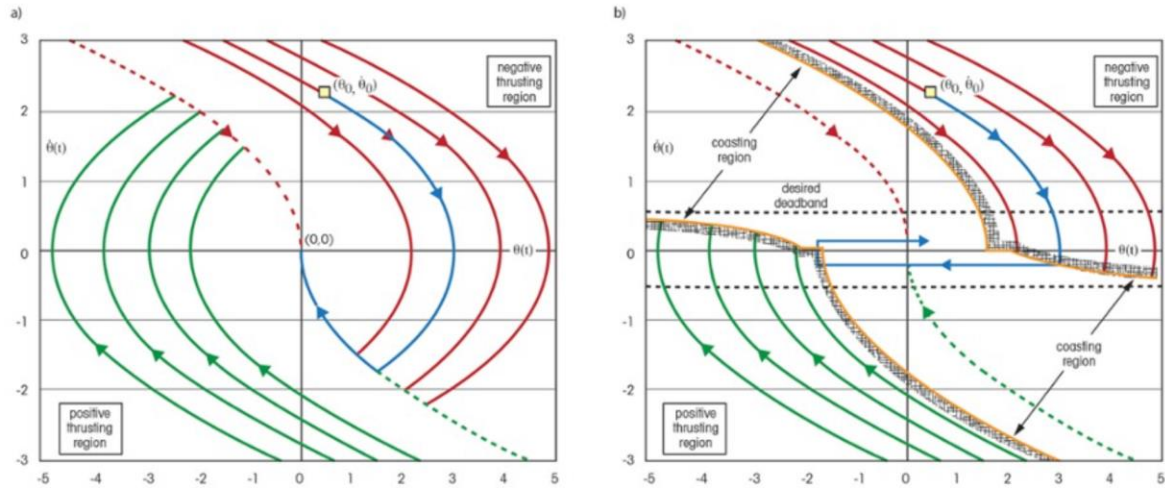


FIGURE 11.14 Switching Curve Plot for a Bang-Off-Bang Controller. Thrusters are either full-on positive, full-on negative, or off. The horizontal and vertical axes are attitude angle, θ , and angle rate, $\dot{\theta}$, respectively. The numbers are notional only, for purposes of illustration. In part a), the blue curves show the optimal trajectory from the initial conditions to the desired state (the origin), under ideal conditions. Part b) shows the imposition of coasting regions, marked by the gray, cross-hatched boundaries. We see the blue trajectory enter into a limit cycle.

The red curves in Figure 11.14a represent vehicle motion when the thrusters are full-on in the negative direction. The green curves are for full positive. The dashed curves denote those that pass through the desired end-state (the origin). Suppose the vehicle is at an initial angular position and velocity of $(\theta_0, \dot{\theta}_0)$, and we desire to get to zero position and velocity (the origin) with as little propellant as possible. As shown in Figure 11.14a, we fire the thrusters negatively, and follow the blue curve until it intersects the positive-thrust curve that intersects the origin. At that point, we fire positively, ceasing when we reach the origin. Ideally, the vehicle would remain there, with the thrusters turned off. However, due to effects like sensor noise, the thrusters would “chatter” on and off near the origin, wasting propellant.

To alleviate this problem, we allow the vehicle some limited movement about the origin, without firing the thrusters—a “coasting region.” See [Figure 11.14b](#) . We limit the motion by establishing a “desired deadband” range of acceptable angular-velocity values, denoted by the horizontal dashed lines in the figure. To enforce this, we set the boundaries for the coasting region as shown by the cross-hatched boundaries. Within those boundaries, the vehicle is allowed to coast at constant θ . To illustrate this, once again assume the vehicle is at initial conditions $(\theta_0, \dot{\theta}_0)$, and we wish to drive in to the origin. We fire negatively until encountering the coasting region boundary on the right of the figure, where we cease firing and let the vehicle coast at constant (negative) θ , until hitting the other boundary, where we fire briefly in the positive direction, until once again entering the coasting region, this time with a positive θ . The vehicle then continues in a “limit cycle,” acceptably close to the origin, and without wasteful chattering of the thrusters.

State Estimation—Kalman Filter

Accuracy requirements may dictate that we use multiple sensors, with different degrees of accuracy, to estimate a single state variable. A common technique for combining such measurements into a single, higher-quality estimate is the *Kalman filter*. This is a recursive algorithm that operates at discrete time-steps, to generate a least-square-error estimate of the vehicle’s state at the current time step. It can also give state estimates at past and future time steps. The Kalman filter combines a linearized mathematical model of the vehicle’s dynamics with an algorithm that calculates a weighted average of the sensor readings, to generate a better state estimate than any of the sensors would yield individually.

The weights used in the averaging process are chosen to be “optimal,” in the sense that they minimize the mean-square error of the estimates. Interestingly, the Kalman filter can even provide estimates of state variables that aren’t directly measured. An additional feature of the Kalman filter is that it also gives a measure of the quality (variance) of the state estimate. For more information about Kalman filters and how they are implemented, see Maybeck [1979].

FireSAT Example

Here we present sample calculations for sizing ADCS actuators for FireSAT. The resulting ADCS design, including sensor selections, is summarized in [Table 11.14](#)  and [Table 11.15](#) .

Table 11.14 FireSAT Spacecraft Control Components Selection. A simple, low-cost suite of components fits FireSAT’s needs. (Adapted from SMAD Table 11-15.)

Type	Components	Rationale
Actuation Devices	(4) Reaction wheels (3) Electromagnets	• 3-axis control plus agility for sensor pointing • Wheel desaturation
Sensors	Horizon sensor Sun sensors Magnetometer	• Provide basic pitch and roll reference • Can meet 0.1° accuracy • Lower mass and cost than star sensors • Initially acquire vehicle attitude from unknown orientation • Coarse attitude data • Fine data for yaw • Coarse yaw data

Table 11.15 FireSAT Spacecraft Control Subsystem Characterized. The baseline ADCS components satisfy all mission requirements.
(Adapted from SMAD Table 11-16.)

Components	Type	Mass (kg)	Power (W)	Mounting Considerations
Reaction Wheels	Four, 0.4 N·m·s	8 (2 kg per wheel)	10 to 20	4 arranged in a tetrahedron for redundancy
Electromagnets	3, 10 A·m ² capacity each	2, including current drive electronics	5 to 10	Orthogonal configuration best to reduce cross-coupling
Sun Sensors	4 wide-angle coarse sensors providing 4 π steradian coverage; $\approx$ 0.1° accuracy	< 1 total	0.25	Free of viewing obstructions and reflections
Horizon Sensors	Scanning type (2) plus electronics; 0.1° accuracy	5 total	10	Unobstructed view of Earth's horizon

Magnetometer	3-axis	< 1 total	5	Need to isolate magnetometer from electromagnets, either physically or by duty cycling the magnets
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☀ Assumed spacecraft characteristics:

- ~ Mass = 174 kg $I_x = I_z$ (I_{yaw}) = 75 kg · m² I_y = 45 kg · m²
- ~ Spacecraft cross-sectional area A_s = 2 m²
- ~ Coefficient of drag C_d = 2.5
- ~ Atmospheric density at 700 km altitude = 1.47×10^{-13} kg/m³
- ~ Difference between center of aerodynamic pressure c_{ps} and center of mass c_m = 0.3 m
- ~ Coefficient of reflectivity q = 0.6
- ~ Difference between center of solar pressure and center of mass = 0.3 m
- ~ Spacecraft magnetic dipole D = 1 A · m²

☀ Requirements:

- ~ Pointing accuracy = 0.1° Slew maneuver = 30° in 10 min

First calculate magnitudes of disturbance torques (initial estimates):

Gravity gradient:

$$T_g = \frac{3\mu}{2R^3} |I_z - I_y| \sin 2\theta = \frac{3(3.986 \times 10^{14})}{2(7.078 \times 10^6)^3} (30) \sin 2\theta$$

$$= 1.77 \times 10^{-6} \text{ N}\cdot\text{m} \quad (\text{assume } \theta = 1^\circ \text{ for normal mode})$$

$$= 4.38 \times 10^{-5} \text{ N}\cdot\text{m} \quad (\text{assume } \theta = 30^\circ - \text{target-of-opportunity mode})$$

$$= 30^\circ - \text{target-of-opportunity mode}$$

Solar radiation:

$$T_{sp} = \frac{F_s A_s (1+q) \cos i}{c} (c_{ps} - c_m) = \frac{1367(2)(1.6) \cos i}{3 \times 10^8} (0.3) \\ = 4.38 \times 10^{-6} \text{ N}\cdot\text{m} \quad (\text{assume worst-case incidence angle of } 0^\circ)$$

F_s is the solar constant, 1367 W/m²

Magnetic field:

$$T_m = \frac{2DM}{R^3} = \frac{2(1)(7.96 \times 10^{15})}{(7078000)^3} = 4.49 \times 10^{-5} \text{ N}\cdot\text{m}$$

(assume worst-case polar magnetic field)

Aerodynamic:

$$T_a = 0.5 \rho C_D A v^2 (c_{pa} - c_m) = 0.5 (1.47 \times 10^{-13}) (2.5) (2) \\ (7504.3)^2 (0.3) = 6.21 \times 10^{-6} \text{ N}\cdot\text{m} \quad (\text{assume worst-case} \\ \text{density } \rho \text{ at } h = 700 \text{ km, which is } 1.47 \times 10^{-13} \text{ kg/m}^3)$$

Total disturbance torque (root-sum square) = $T_{D \text{ total}} = 6.3 \times 10^{-5} \text{ N} \cdot \text{m}$

Consider Reaction Wheels. Size them for both torque capability and momentum-storage capacity.

For torque capability, consider three situations. First, find the moment required for peak disturbance rejection. For worst case, assume one wheel rejects all of $T_{D \text{ total}}$. Assume a margin factor of 0.2. The moment, M_{RWD} , is given by

$$M_{RWD} = T_{D total} (1 + \text{margin factor})$$

$$= 6.3 \times 10^{-5} \text{ N}\cdot\text{m} * 1.2 = 7.6 \times 10^{-5} \text{ N}\cdot\text{m}$$

Now calculate the moment, $M_{RW \text{ slew}}$, required for a slew maneuver (30° in 10 minutes, as per requirements). A “maximum-acceleration slew” would reach half the desired slew angle in half the time. To implement this, first integrate the equation of motion for a rotating body,

$$\theta = \frac{1}{2} \ddot{\theta} t^2$$

Then, for our maximum-acceleration slew,

$$\frac{\theta_{\text{slew}}}{2} = \left(\frac{1}{2} \right) \frac{M_{RW \text{ slew}}}{I} \left(\frac{t_{\text{slew}}}{2} \right)^2$$

$$M_{RW \text{ slew}} = 4 \theta_{\text{slew}} I / t_{\text{slew}}^2 = 4 \times (30^\circ \times \pi \text{ rad} / 180^\circ) \times 75 \text{ kg} \cdot \text{m}^2 / (600 \text{ s})^2$$

$$= 4.4 \times 10^{-4} \text{ N} \cdot \text{m}$$

Since $M_{RW \text{ slew}} > M_{RWD}$, you should use the slewing moment for sizing the wheels. To be very conservative, size them to provide $4 \times 10^{-3} \text{ N}\cdot\text{m}$ torque, which is nine times the maximum expected slew torque.

Now size the reaction wheels’ angular momentum capacity, per wheel. As with the moment requirement, consider both disturbance-rejection and slew maneuvers. For disturbance-rejection, integrate the worst-case cyclic disturbance over half an orbit—or one quarter of an orbit, if magnetic disturbance is the largest). Use T_m , since it’s the greatest of the cyclic disturbance torques. The angular momentum, H_D , gained due to disturbance rejection is then:

$$H_D = 0.707 T_M [P / 4]$$

where P is the orbital period, in seconds.

$$H_D = 0.707 \times 4.49 \times 10^{-5} \text{ N}\cdot\text{m} \times (5917 \text{ s})/4 = 0.047 \text{ N}\cdot\text{m}\cdot\text{s}$$

Now estimate the wheels' maximum angular momentum change during a slew maneuver. For the slew maneuver described above, the wheel accelerates the vehicle during the first half of the maneuver, and then decelerates it for the second half. So, the maximum change in wheel angular momentum, H_{slew} , occurs at the halfway point, or 300 seconds.

$$H_{slew} = M_{RW\ slew} \Delta t = 4.4 \times 10^{-4} \text{ N}\cdot\text{m} \times 300 \text{ s} = 0.132 \text{ N}\cdot\text{m}\cdot\text{s}$$

Since $H_{slew} > H_D$, you should size the wheel angular momentum storage capacity based on the slewing requirement. To be conservative, select wheels of 0.4 N·m·s capacity each, which gives a safety factor of 3.

Thus, the reaction wheels are sized for up to 4×10^{-3} N·m torque and 0.4 N·m·s angular momentum capacity. [Table 11.12](#)  shows that both values are reasonable for typical small-satellite applications. Use three mutually perpendicular wheels, plus one more wheel for redundancy.

Use Magnetic Torquers for Momentum Dumping. As a worst case, assume only one of the magnetic torquers will dump all the momentum of a saturated reaction wheel. Earth's magnetic field varies considerably over any single orbit, but as a rough approximation, assume that the magnetic torquer provides a constant moment, M_m , over 1/5 of the orbit, and no moment over the rest of the orbit. Furthermore, assume the torquer will dump all the angular momentum during just one orbit.

Define H_{dump} to be the total angular momentum to be dumped, which is simply the wheel's maximum capacity. From the wheel-sizing calculations above,

$$H_{dump} = 0.4 \text{ N}\cdot\text{m}\cdot\text{s}$$

Dumping this momentum by turning on the magnetic torquer for 1/5 of an orbit gives

$$H_{dump} = \int_0^{0.2P} M_m dt$$

Where, again, P is the orbital period. Since M_m is constant over that part of the orbit, integrate and rearrange to get

$$M_m = H_{dump}/(0.2P)$$

NIA is the quantity we're seeking, the capacity of the magnetic torquer. Now we have

$$M_m = |\mathbf{D} \times \mathbf{B}| = NIA \cdot B$$

NIA is the quantity we're seeking, the capacity of the magnetic torquer. Now we have

$$NIA = \frac{H_{amp}}{0.2P \cdot B}$$

The orbit's inclination of 55° is between those of a polar orbit and an equatorial orbit, so use [Table 11.10](#) and estimate B as 0.75 times the polar orbit value, which in turn is $2M/R^3$, where M , Earth's magnetic moment = $7.96 \times 10^{15} \text{ tesla} \cdot \text{m}^3$. For the FireSAT mission, the orbit radius, R , is 7,078,000 m, and $P = 5917 \text{ s}$. So we now have

$$NIA = \frac{0.4N \cdot m \cdot s}{0.2 \cdot 5917s \cdot 0.75 \times 2 \times (7.96 \times 10^{15} \text{ tesla} \cdot \text{m}^3 (7,078,000\text{m})^3)}$$

or

$$NIA = 10 \text{ N} \cdot \text{m}$$

$$NIA = 10N \cdot \frac{m}{\text{tesla}} = 10A \cdot m^2$$

To be conservative, use torque rods with $30 \text{ A} \cdot \text{m}^2$ capacity each (safety factor of 3). [Table 11.12](#) shows that this is a reasonable value for small-satellite applications. Use three mutually-perpendicular rods and one redundant rod for the most critical direction.